07/02/24

Algorithm (PB)

Counting Sort (Non comparison based sort)

↳ inputs are in a range

~~↳ comparison based sorting~~

What is the lower bound for comparison based sorting: $\Omega(n \log n)$

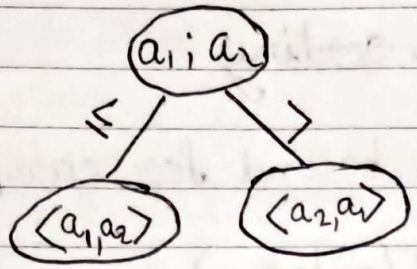
<u>Theorem</u>: Any comparison based sort Algorithm requires $\Omega(n \log n)$ comparisons in the worst ~~cost~~ case. ($h \leq cn \log n$) (only heap and merge sort, called optimal sort.)

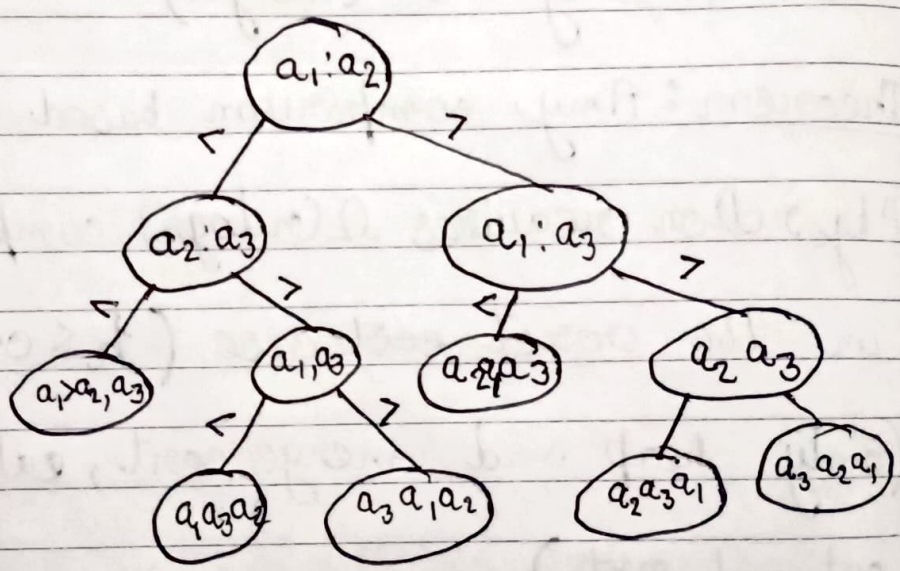
Pf consider a binary decision tree model

<u>Full</u> binary decision tree

↳ 0 or 2 children for every node

decision tree represent comparison between elements that are performed by a sorting algo on a given input size n. a_1, a_2





there are all possible permutations in the final nodes which is n!

obs (1) all n! permutations are present as leaves.

② number of comparisons is height of the decision tree.

consider height of decision tree is h and let l be the length from root to a leaf. Let L be the no of leaves reachable from the root.

$n! < L < 2^h \Rightarrow$ # of leaves atmost.

$\Downarrow$

no of permutation that are leaves in decision tree.

$h = \log$ on both sides of (1)

so

$h \geq \log(n!)$

$h \geq \sum_{i=1}^{n} \log i$

$\left(\log 1 + \log 2 + \ldots + \log n = n/2 \log n/2 \right)$

$h \leq n/2 \log n/2$

$= \Omega(n \log n)$ (Proved)

Divide and Conquer:

① Binary Search:

$$T(n) = T(n/2) + 1 = \Theta(\log n)$$

observation: We discard 1/2 elements during every recursion.

- **Median Finding Algorithm**

Naive Finding Algo

Naive algo: sorting & find the $n/2$ th element: $\Theta(n \log n)$

using D&C $\Rightarrow \Theta(n)$

→ Find the kth smallest element from an input of n elements :

~~↳ Algorithm~~

We can find if a number is median ~~by~~ in $O(n)$ time ~~the~~ by seeing if

the number of elements smaller than the number is $n/2$.

Algorithm to find meadian (~~Select~~ Algorithm is called "select")

Select(A, p, r, k) -> return k^{th} smallest element from A [p ... r]

if p == r return A[p]

else{

x = choosepivot (A, p, r)

q = Partition(A, p, r, x) //A[p...q-1] A[q+1..r]

xRank = q - p + 1 //Position of partition

//the pivot is k^{th} smallest

if (k = xRank) return x

~~else~~

else if $k < xRank$

return Select(A, p, q-1, k)

else

return Select(A, q+1, r, k-xRank)

<u>Choose Pivot</u>

- Given n inputs, make g groups, each of size 5 elements $\Rightarrow \lfloor \frac{n}{5} \rfloor$ groups
- Sort each group if size 5 in constant time, take median of each group.

Let $M = \{g_1^m, g_2^m, \ldots, g_{n/5}^m\}$ median

- Use select(M, 1, n/5, k'), where k' is median of n/5 groups. to get median of medians: mm

Example: 8, 33, 17, 51, 57 | 49, 35, 11, 25, 37 | 14, 3, 2, 13, 52 | 12 6 29 32 54 | 5, 16, 22, 23, 7

② Sort the groups

8, 17, 33, 51, 57 | 11, 25, 35, 37, 49 | 2, 3, 13, 14, 52 | 6, 12, 29, 32, 54 | 5, 7, 16, 22, 23

③ M = {33, 35, 13, 29, 16}, find mm select

mm = 29 Pivot

A< = {8, 17, 11, 25, 14, 3, 2, 13, 12, 6, 5, 16, 22, 23, 7}

|A<| = 15

A> = {...} |A>| = 9

② Select (A<, 1, 15, 13)

~~A< =~~ A< = {8, 17, 11, 25, 14 | 3, 2, 13, 12, 6 | 5, 16, 22, 23, 7}

M = {14, 6, 16} mm = 14

A< = {8, 11, 3, 2, 13, 12, 6, 5, 7}

A> = {16, 17, 25, 22, 23} |A>| = 5

=> select (A<, 1, 5, 13-10)

$mm = 22$

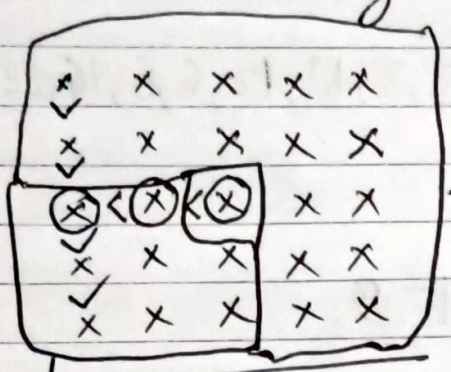
return 22 Kth smallest element

time complexity Analysis

$$T(n) = T(n/5) + T(?) + O(n)$$

$\downarrow 3n/4$

Worst case analysis



$\rightarrow 3n/4$

$\Rightarrow n - 3n/4 \times n/5$

$\Rightarrow n - 3n/10$

$\Rightarrow 7n/10$ (# of elem $> mm$)

$n/5$ groups

$\rightarrow$ smaller than mm (medain of median)

Assume an algo discards 1/3 element during each recursion

$$cn + 2/3\,cn + (2/3)^2 cn + \dots + (2/3)^k cn$$

1/3 of 2/3 is discarded $\Rightarrow$ 2/9 cn is discarded

$\Rightarrow 2/3\,cn - 2/9\,cn = 4/9\,cn$

it is a gp series $\sum_{j=0}^{\infty} cn(2/3)^j = cn\frac{1}{1-2/3} = 3cn$
$\hookrightarrow <1$
$= \Theta(n)$

$T(n) = T(c_1) + T(c_2) + O(n)$
$\hookrightarrow O(n)$

and $\underbrace{c_1 + c_2 < n}$
$\hookrightarrow$ discarding a fraction of n during recursion

$$\left.\begin{aligned} n/5 + 3n/4 &= \frac{19}{20}n < n \\ n/5 + 7n/10 &= 9n/10 < n \end{aligned}\right\}$$

Th^m: $T(n) \leq cn$ or $T(n)$ is $\Theta(n)$

if $n=1$ $T(n) \leq cn$ as long as $c \geq 1$

Assume $T(n') \leq cn'$ for all $n' < n$. To show $T(n) \leq cn$

We have $T(n) = T(n/5) + T(3n/4) + n$

$\because$ $n/5$ & $3n/4$ are less than n, apply induction hypothesis

$T(n) \leq cn/5 + c\,3n/4 + n = cn(1/5 + 3/4) + n$

$= n\left(\frac{19c}{20} + 1\right)$

$T(n) \leq cn$ provided we select c such that $c \geq \left(\frac{19c}{20}\right) + 1$

$\Rightarrow c \geq 20$

$c \geq 1$ & $c \geq 20$ the condition $T(n) \leq cn$ holds and thus $T(n)$ is $\Theta(n)$

Why group of 5?

~~$T(n) = T(n/3$~~